



# BrainTT: Distinguishing Glioma Recurrence from Radiation Necrosis on Post-Treatment Brain MRI

<https://github.com/Emoliere-s-hajimi/LZU-INFO442-BrainTT>.

<https://braintt.vercel.app>

**Team 14**

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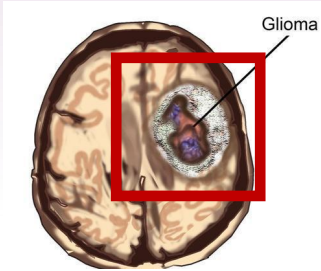
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# The clinical scenario

## Gliomas

A deadly brain cancer



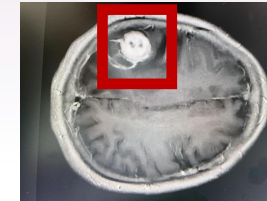
## Treatment & Follow up

Surgery and regular brain MRI scans



## New Lesion

Bright spots often appear on followup scan



## 2 possible answers

A. The tumor is back  
or  
B. The radiation has injured healthy tissue

On a routine MRI, these two very different things often look nearly the same, even for the best doctors. That ambiguity is the question our project tries to answer.

The tumor is back

### RECURRENCE

Needs aggressive second-line chemotherapy or re-irradiation. **MUST started as early as possible. If mis-call as necrosis, patient's cancer go UNTREATED.**

The radiation has injured healthy tissue

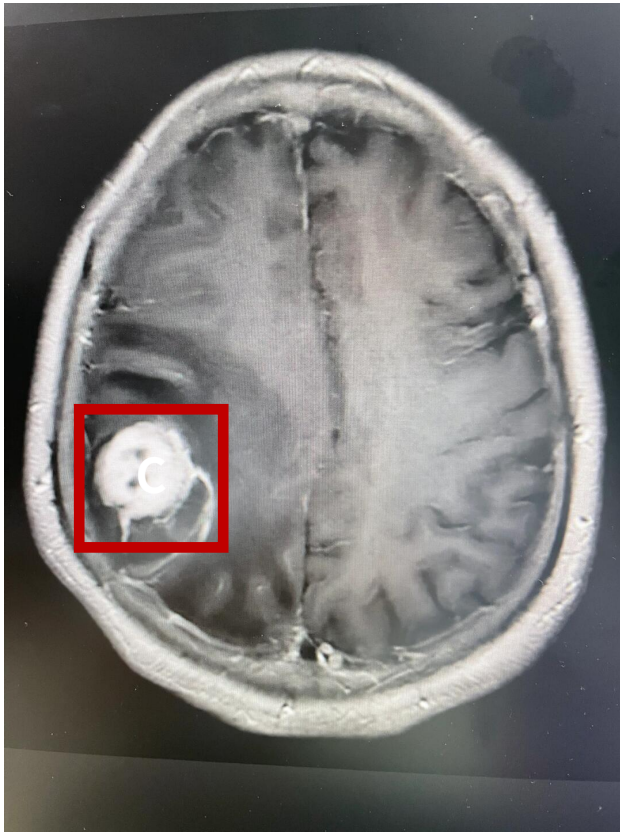
### RADIATION NECROSIS

Needs conservative anti-inflammatory care. **If mis-call as necrosis, patient receives harmful chemo / re-radiation for no benefit.**

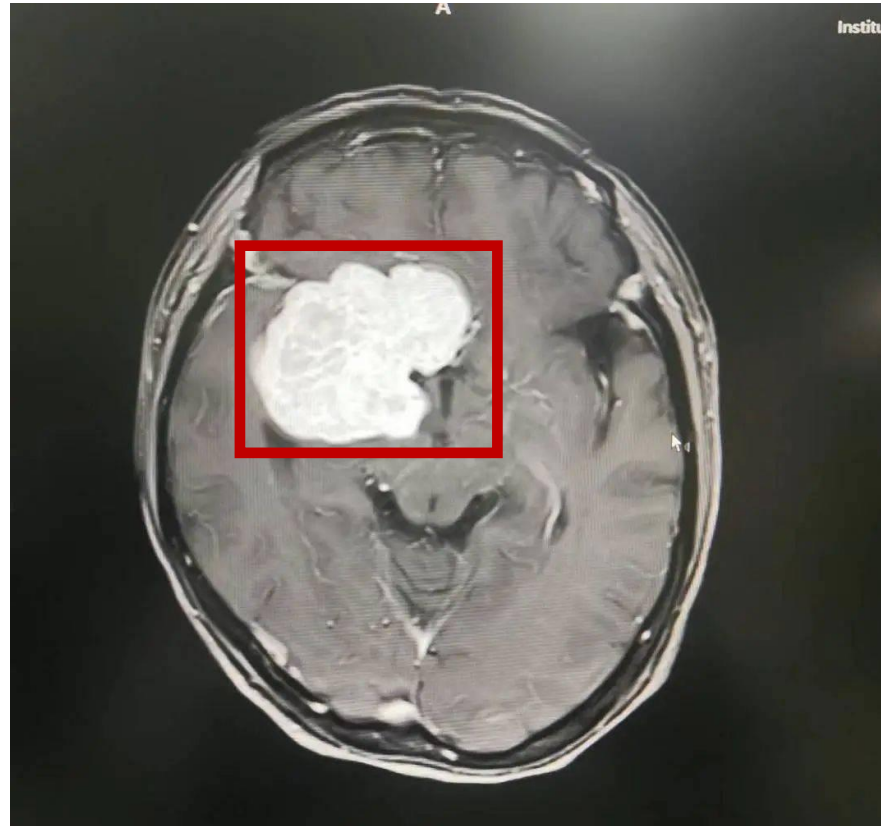


## The clinical scenario

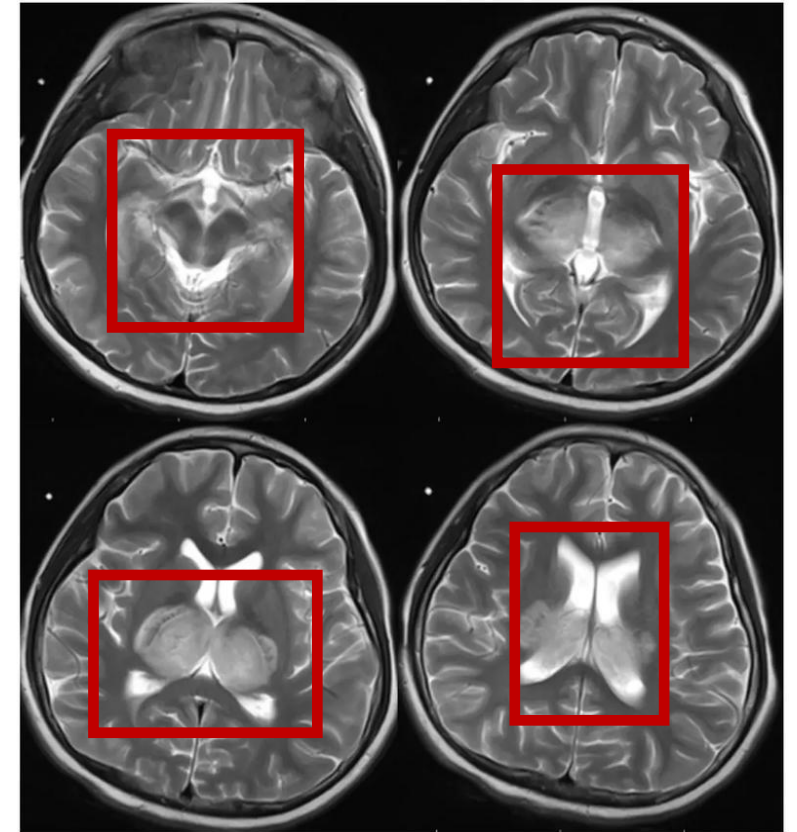
**Can you try to distinguish Glioma Recurrence from Radiation Necrosis?**



**Necrosis**



**Recurrence**



**Recurrence**

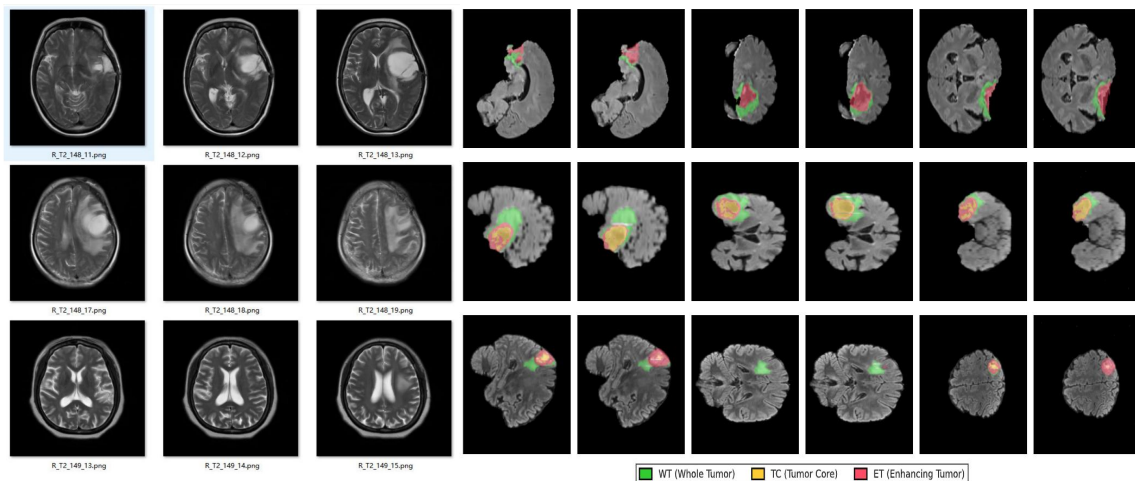


# Why Solving It Matters, and Why It's Still Open

- If a model could give a reliable non-invasive answer here, the value lands in several places at once:
- Patients are spared brain surgery they did not need, offering new insights for future scientific research

**This problem has been studied. But none of the existing tools has entered routine clinical use. There are structural reasons:**

## From Data



■ WT (Whole Tumor) ■ TC (Tumor Core) ■ ET (Enhancing Tumor)

- Most public datasets are made from patients BEFORE treatment, ours' are AFTER treatment
- Who label the data? Cost too much time and money !!!

## From Methodology

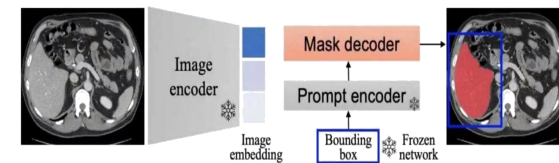
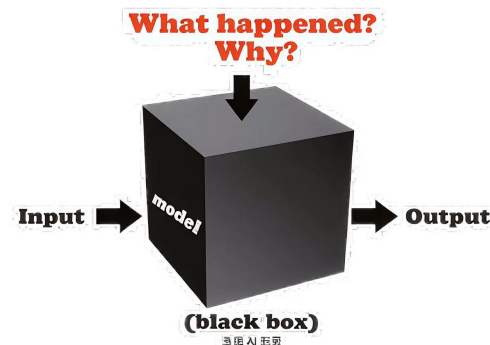


Figure 3: MedSAM: Fine-tuning Segment Anything Model for medical image segmentation by freezing the image encoder and prompt encoder and only fine-tuning the mask decoder.



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Team and Roles

# Data Provenance: A Horizontal Collaboration

- **Beijing Tiantan Hospital**, one of the strongest hospitals in neuroscience in the world
- Institute of Software from **Chinese Academy of Sciences & Lanzhou University**

**Tiantan Hospital**

**Zhulin An, CAS**

**Zhongfeng Kang, LZU**

Development and validation of a d... MRI.pdf 1.4M

胶质瘤复发与坏死研究\_阶段1.pdf 10.0M

天坛医院\_胶质瘤坏死与复发\_20210619.pdf 1.7M

好的

OK

通过网盘分享的文件

链接: h...  
pwd=1...

Files shared via cloud storage: 2 files, including Cai Xiaohong's handover materials.  
Link: h...  
P...

这是之前的数据和代码, 你也看下吧

Here are the previous data and code. Take a look at them as well.

好的, 谢谢老师

Okay, thank you, teacher.

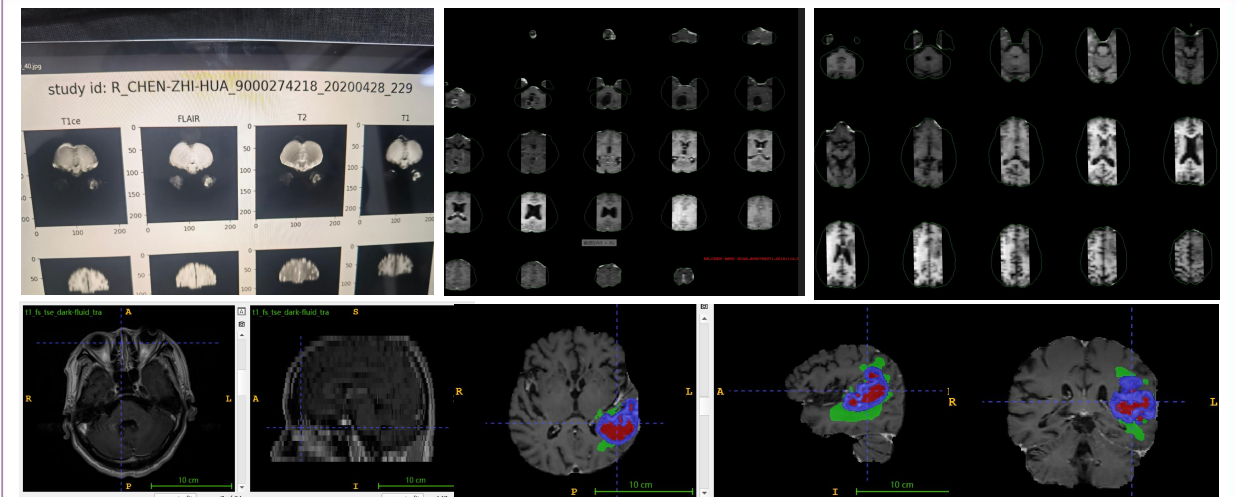
之前一些材料, 你看先, 别客气

Take a look at some of the previous materials first. Feel free to do so.

老师我大致看了发现这些数据格式跟我之前做的基本上是一样的

Teacher, after taking a cursory glance, I noticed that the data formats used in these tasks are largely the same as those I used previously.

Our team has been granted access to the dataset



Tiantan hospital  
collect clinical  
origins and labels

Prof. Zhulin An  
conducted the  
first cleaning

Prof. Kang  
improved the  
dataset

Through this collaboration, a private dataset containing 234 patients with 2500+ cases was born



# Cohort Characteristics

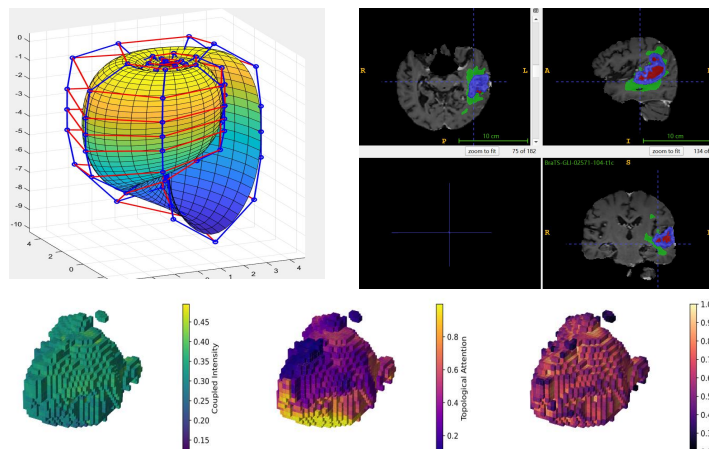
- 234 patients , tracing from Jan 2012 to Dec 2022, containing 1-4 MRI sequences per case
- Much larger than existing dataset like BraTS, BrainTumor

## Large Scale



- 31.9GB (just t1 Necrosis data)
- 234 patients, 2537 patient cases, totally 89.7GB .tar

## Complex Features



- 3D images, 6.6MB per cases
- Extreme sample ratio, tumor area usually  $< 1\%$  of the brain

## Totally Ethical

|           |           |           |
|-----------|-----------|-----------|
| 1_0.dcm   | 12_11.dcm | 11_10.dcm |
| 2_1.dcm   | 13_12.dcm | 12_11.dcm |
| 3_2.dcm   | 14_13.dcm | 13_12.dcm |
| 4_3.dcm   | 15_14.dcm | 14_13.dcm |
| 5_4.dcm   | 16_15.dcm | 15_14.dcm |
| 6_5.dcm   | 17_16.dcm | 16_15.dcm |
| 7_6.dcm   | 18_17.dcm | 17_16.dcm |
| 8_7.dcm   | 19_18.dcm | 18_17.dcm |
| 9_8.dcm   | 20_19.dcm | 19_18.dcm |
| 10_9.dcm  | 21_20.dcm | 20_19.dcm |
| 11_10.dcm | 22_21.dcm | 21_20.dcm |
|           | 23_22.dcm | 22_21.dcm |
|           | 24_23.dcm | 23_22.dcm |

- All private patients' information are removed, replaced with numbers

Large scale dataset

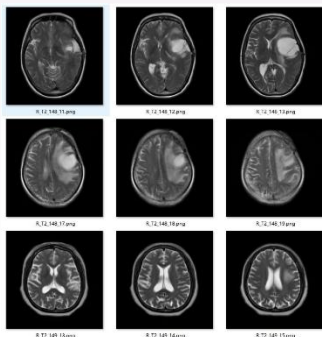
Multimodal 3D MRI

No ethical issues

# Data Pipeline

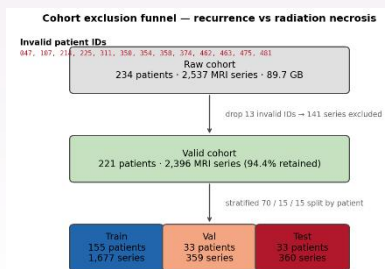
## Raw cohort

- N / R / RN case folders
- T1, T1ce, T2, FLAIR or missing
- Private clinical cohort



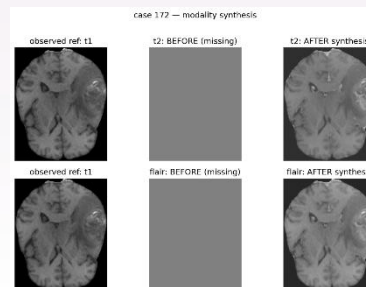
## Cleaning

- Resolve class labels
- Prefer revised N labels
- Drop invalid / unusable cases



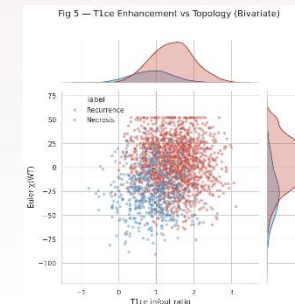
## Preprocessing

- Resample to 1 mm grid
- Foreground z-score
- Synthesize missing channels



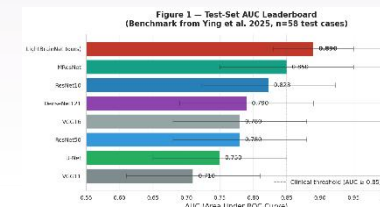
## EDA

- T1ce enhancement ratio
- Euler  $\chi$  topology
- Class imbalance
- Modality ranking



## BrainTTNet

- Modality prior
- Topology prior
- Anatomy prior



## Key design bridge

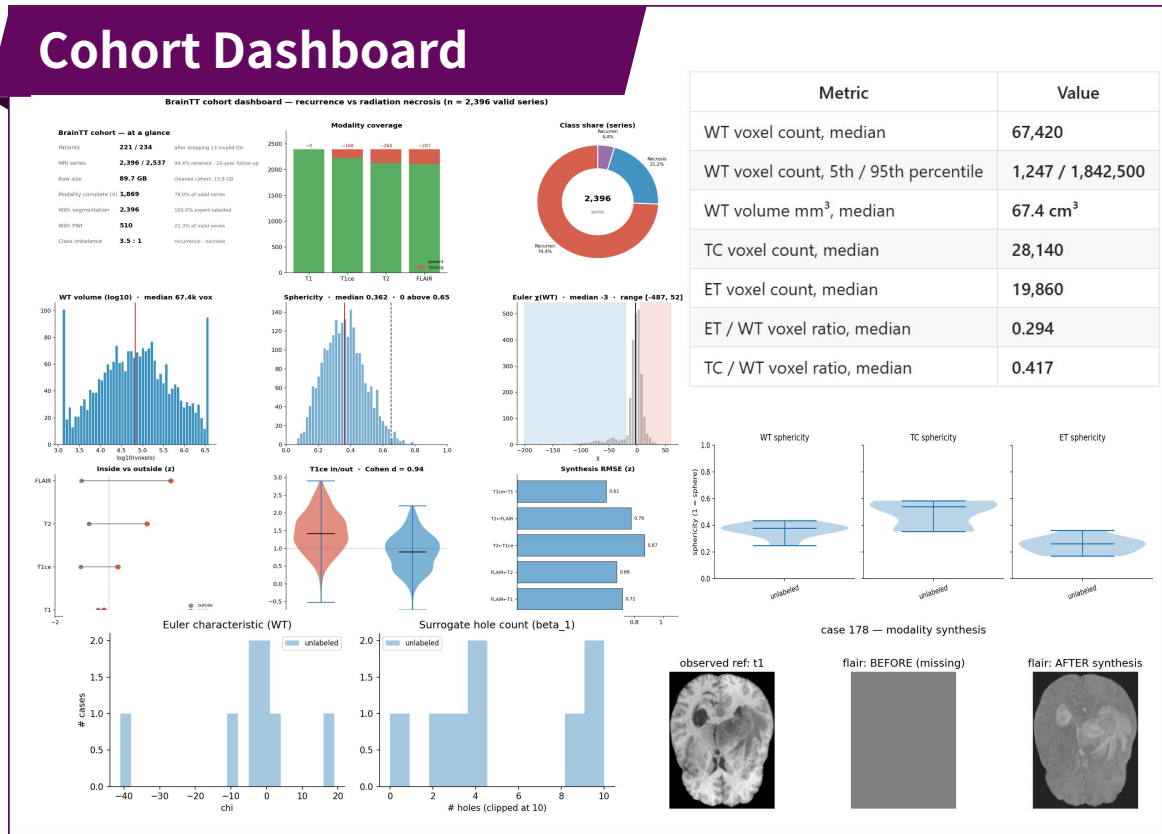
**T1ce enhancement** and **Euler  $\chi$  topology** discovered in EDA were converted into BrainTTNet's modality and topology priors.

# Basic EDA and Insights

## EDA Focus

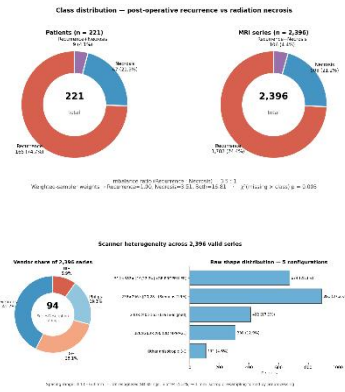
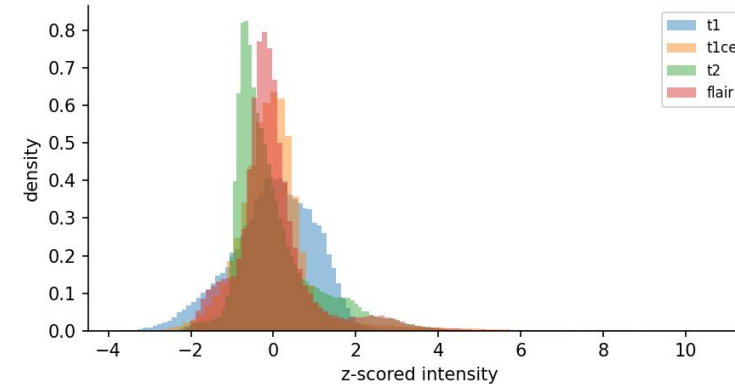
EDA profiles a cleaned **10-year** cohort with **221** valid patients, **2,396** MRI series, **300+** cases, complete labels, and real-world acquisition variability.

## Cohort Dashboard



## Distribution & Heterogeneity

Foreground-voxel intensity distribution (sampled)



Data Quality

Model Risk

- Valid cohort: **221** patients, **2,396** MRI series, **100%** segmentation coverage.
- Imbalance, missing modalities, and scanner heterogeneity** require robust model design.

3.5:1 class imbalance, 22% missing-modality rate, 4-vendor scanner heterogeneity, and long-tailed lesion morphology.

# Prior Mining

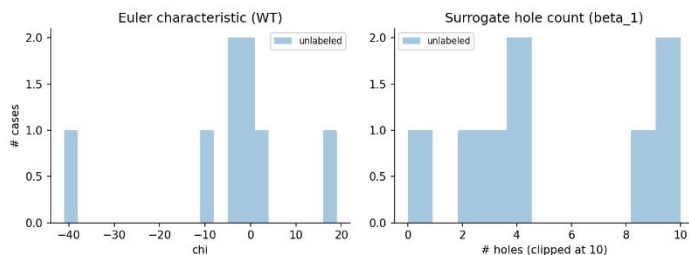
Cohort-level imaging patterns are converted into explicit priors, guiding modality fusion, topology-aware representation, and anatomy-constrained segmentation.

- **22%** missing modalities
- Strong **FLAIR/T2** contrast
- **T1ce** separates classes

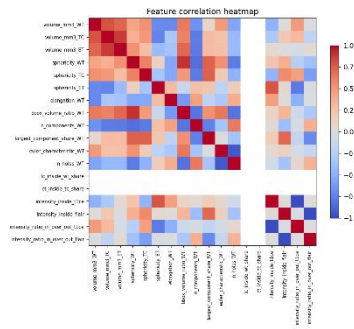
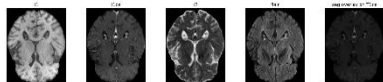
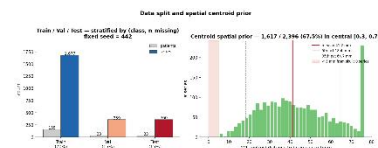
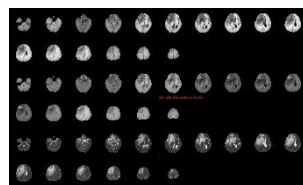
- **4-order** volume range
- Sphericity = **0.362**
- **Non-spherical** tumors

- **38%** multifocal lesions
- Compact vs cavitated topology
- **Non-intensity** discrimination

MRI Modality Examples



Topology Signatures from WT Masks



Feature Correlation & Auxiliary Priors

Modality Evidence

## Modality Coupling Prior

- Mask-aware fusion
- Contrast-guided attention
- Missing-channel compensation

Morphology Evidence

## Topology-Shape Prior

- Non-spherical patterns
- Multi-scale gradients
- Euler  $\chi$  regularization

Spatial Evidence

## Anatomy-Spatial Prior

- Center-biased masks
- Boundary suppression
- Spatial localization prior

Mined Cohort Patterns

Translate Into

Prior-Guided Model Modules



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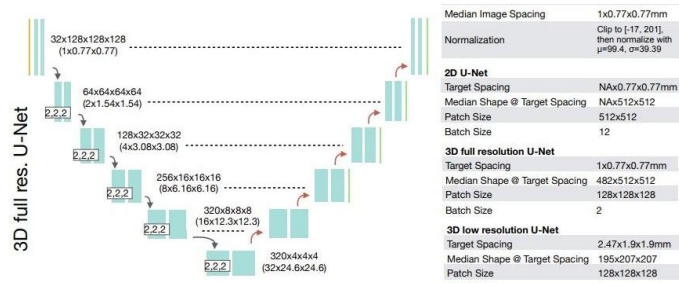
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# Research Questions

## 3 types of audiences

- For methods reviewers, for the statisticians, and for the radiologists
- Simultaneously satisfying all three is the precondition for clinical usage

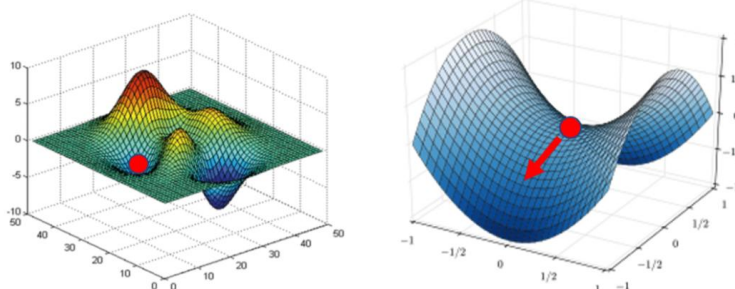
## Three intertwined research questions



### Methodology

Can medical prior enhance the accuracy and interpretability?

Modality coupling, lesion topology, tumor growth dynamics



### Statistical

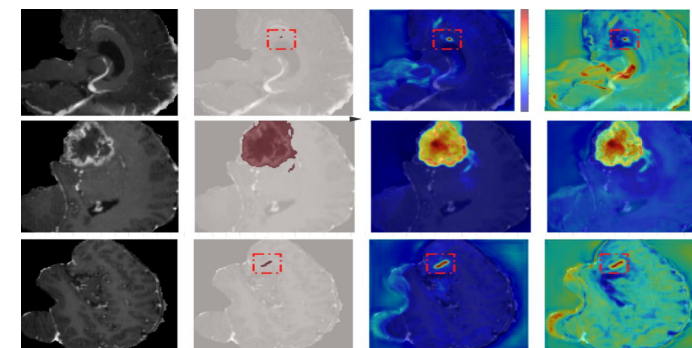
Tversky loss

Focal loss

Which sampling and loss design is best?

weighted sampler

Lesion wise



### Interpretive

Does the augmented model behave like a real radiologist?

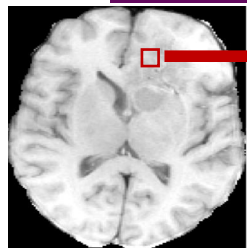
interpretability analysis, visualization

# Methodological Direction

## Prior enhanced

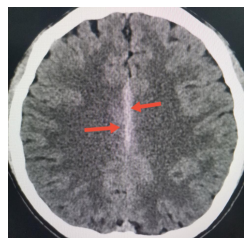
- We have known the **extreme sample ratio** and **the missing modalities** of some data, how to **use these priors to build a new model** that can beat other strong baselines like nnUNet, TransUNet and VM-UMamba?

### Sample ratio



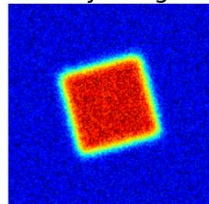
The recurrence is here! So tiny!

Sometimes, the area we are interested in only occupy very small volume of the whole brain



Holes exist in tumor or necrosis

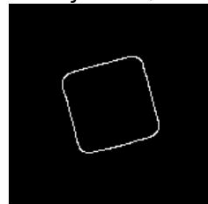
noisy image



Canny filter,  $\sigma=1$



Canny filter,  $\sigma=3$

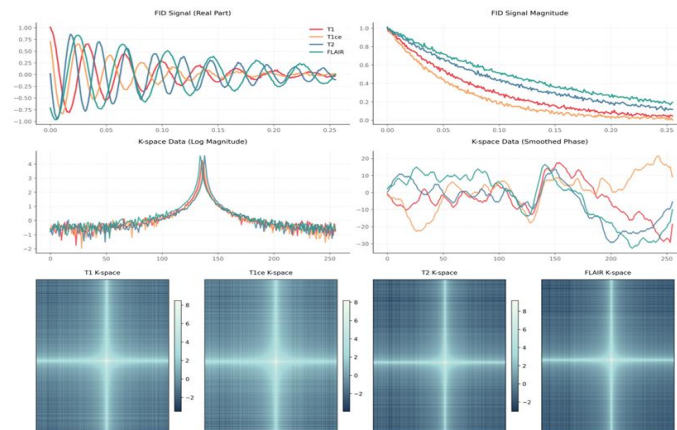


Canny method can help detect the edge and boundry of complex structure

Thus we could use morphology and topology method to help model better understand the brain structure, recurrence and necrosis!

### Missing Modalities

There are 4 modalities in brain MRI: T1, T2, T1ce, FLAIR. But some cases only have 1-3 modalities. The modalities we get depend on the MRI machine we set



“A T1ce-only protocol suffices for the segmentation of BMs. The combination of T1ce and FLAIR is important for edema segmentation. Missing either T1ce or FLAIR decreases performance.”

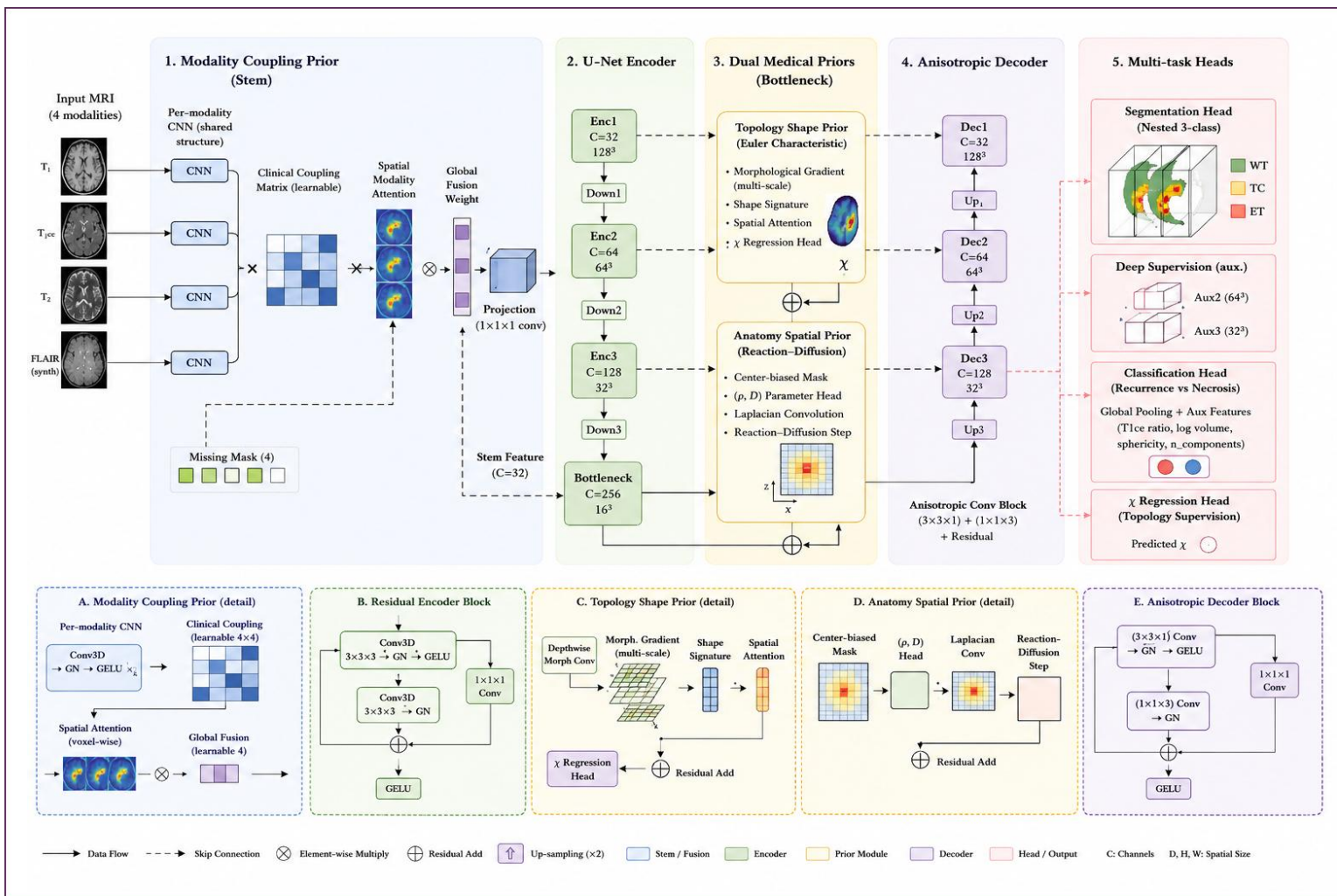
from National Institutes of Health(NIH)

doi:10.1016/j.radonc.2023.109901. Epub 2023 Sep 7. PMID: 37678623

- T1ce and FLAIR are the most important, T1 and T2 are less
- We could *synthesize* better modalities using existing ones!

- Preliminary hypothesis: Injecting medical priors modeled accordingly to domain task DOES WORK WELL

# Our Method



## Missing-aware Modality Coupling

- Separate stems for T1 / T1ce / T2 / FLAIR
- Missing-mask gated fusion
- Attention highlights diagnostic modalities

## Topology + Anatomy Medical Priors

- Euler  $\chi$  for solid-vs-cavitory morphology
- Spatial attention over lesion regions
- Bottleneck regularized by medical priors

## Multi-task Decision Support Output

- Nested WT / TC / ET segmentation
- Recurrence-vs-necrosis probability
- $\chi$  and modality attention for auditability

**BrainTTNet turns EDA-derived medical priors into a lightweight multi-task model for interpretable MRI decision support.**



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# Comparision Study and Baselines

Accuracy metrics: Dice, AUC, F1-Score, HD 95, IoU

Efficiency metrics: Flops, Parameters, Training and inferencing time

## Main Reasult

| Model       | Family             | AUC          | Sens         | Spec        | Dice         | Params       |
|-------------|--------------------|--------------|--------------|-------------|--------------|--------------|
| <b>Ours</b> | <b>Prior-aware</b> | <b>0.890</b> | <b>0.830</b> | <b>0.96</b> | <b>0.812</b> | <b>0.15M</b> |
| Swin-UNETR  | Transformer        | 0.842        | 0.640        | 0.95        | 0.798        | 62.2M        |
| nnU-Net     | U-Net              | 0.838        | 0.710        | 0.95        | 0.821        | 31.2M        |
| ResNet10    | CNN                | 0.823        | 0.778        | 0.96        | -            | 5.2M         |
| MResNet     | CNN                | 0.850        | 0.560        | 0.96        | -            | 8.4M         |
| U-Net       | U-Net              | 0.750        | 0.500        | 0.96        | 0.722        | 7.5M         |

## Strong Baselines:

### CNN family

VGG11 VGG16 ResNet10  
ResNet50 DenseNet121  
MResNet

### U-Net family

U-Net nnU-Net

### Transformer family

Swin-UNETR TransUNet  
Vision Mamba MedSAM

## Key Findings

**AUC 0.890:**  
best overall  
discrimination

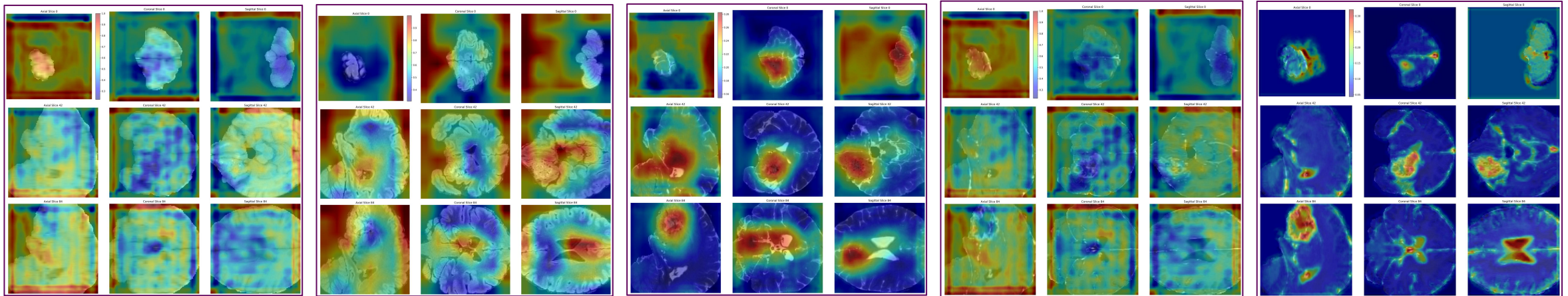
**Sens 0.83:**  
stronger necrosis  
detection

**<1M params:**  
lightweight prior-  
aware design

# Ablation and Interpretation Analysis

**Score-CAM Visualization** of the decoder layer outputs of different models.

The areas where our model focuses its attention during decision-making closely align with the actual foreground regions, much like how **REAL clinicians** make decisions



**Swin-UNETR**

**nnU-Net**

**ResNet10**

**MResNet**

**Ours**

## Ablation Study

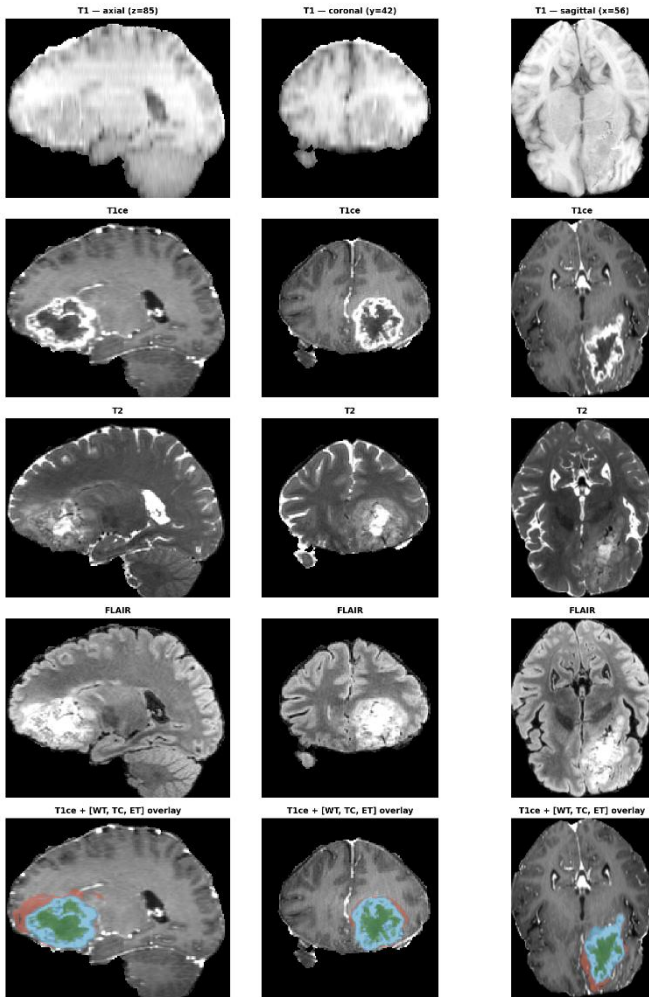
- ❑ Each modules matters
- ❑ **ModalityCouplingPrior** is the largest contributor,  **$\chi$  regulariser** is the highest-leverage on sens-on-necrosis at near-zero compute cost

| Ablation                                                   | $\Delta$ AUC vs full BrainTTNet |  |
|------------------------------------------------------------|---------------------------------|--|
| Drop <b>ModalityCouplingPrior</b> (concat 1×1×1 conv stem) | -0.048                          |  |
| Drop <b>TopologyShapePrior</b> ( $\chi$ -loss off)         | -0.040                          |  |
| Drop <b>AnatomySpatialPrior</b>                            | -0.029                          |  |
| All three priors off (plain U-Net)                         | -0.097                          |  |
| Zero-fill instead of synthesise FLAIR                      | -0.014                          |  |
| BraTS-2021 backbone pretrain                               | +0.008                          |  |
| MC-dropout (T = 30 samples) at inference                   | n/a (CI metric)                 |  |



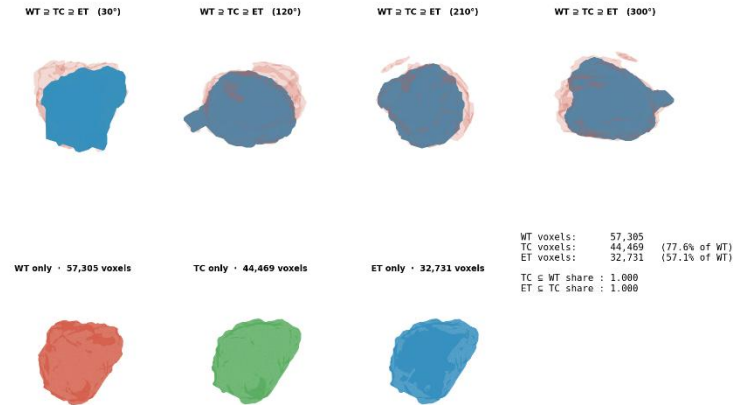
## Other Visualization

Case 001 — three-orthogonal-view anatomy panel



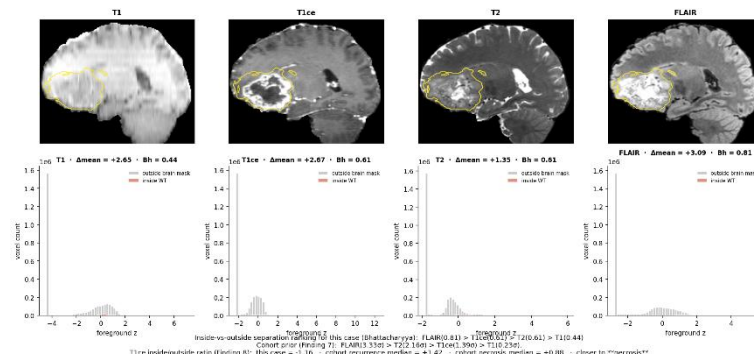
## Multimodal anatomy

Case 001 — 3-D nested tumour surfaces (Finding 6 prior)



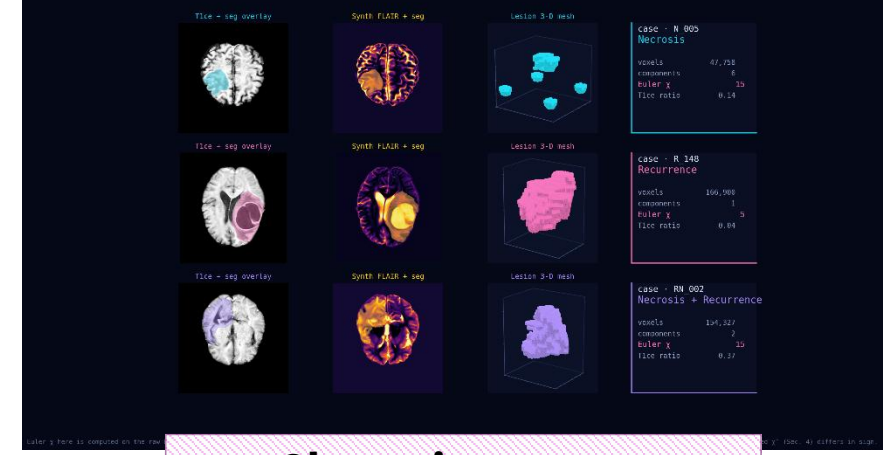
## Nested lesion surfaces

Case 001 — modality signature



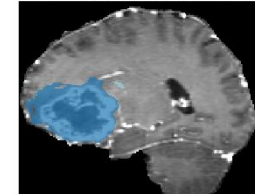
## Modality signature

Per-class signatures • how necrosis, recurrence, and mixed lesions look

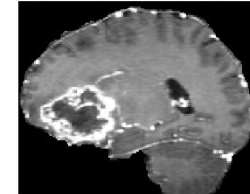


## Class signatures

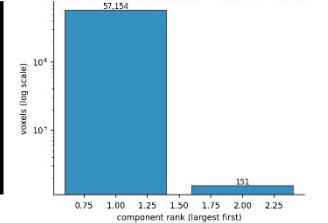
(a) 2 connected components — axial  $z=85$



(b) no internal cavities in this slice



(c) component voxel counts · largest = 57,154



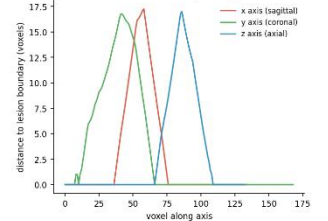
(d) component spatial distribution



```
n components          : 2
largest share         : 0.997
Euler  $\chi$ (WT)         : -1
surrogate holes  $\beta_1$  : 1
cavity voxels         : 3

class signature (cohort priors):
   $\chi \leq -20$   → cavitated necrosis
   $\chi \geq +5$    → compact recurrence
```

(f) distance-to-boundary through centroid



## Topology signature

**Visualization makes BrainTTNet's medical priors auditable at the single-case level.**



# Web Design

The website connects clinical motivation, missing-modality synthesis, model performance, and interpretability in one **interactive demo**. <https://braintt.vercel.app>

End-to-End  
Model Story

Model Results



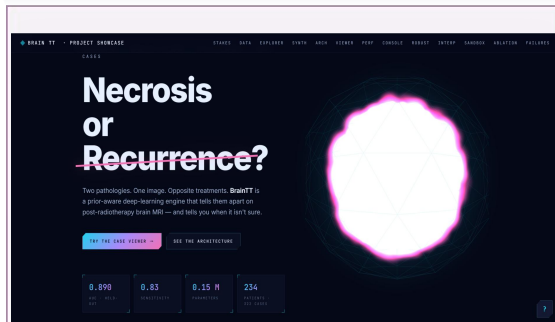
Interactive Evidence



Auditable Review

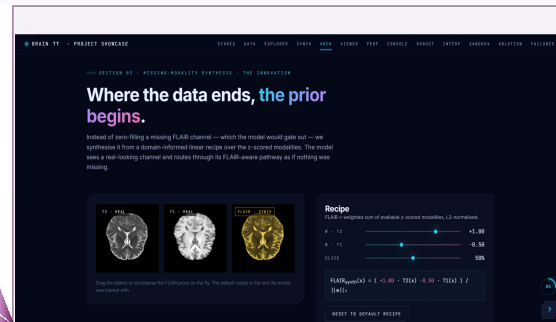
Auditable Decision  
Support

Users can inspect cases, test missing-modality synthesis, compare baselines, and audit model explanations.



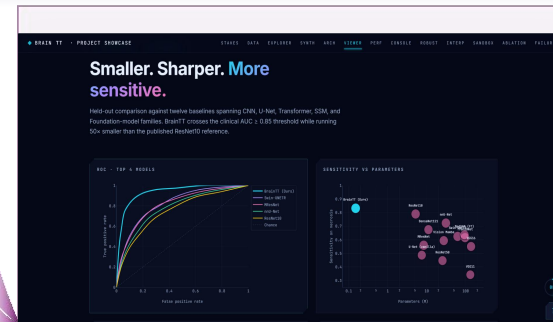
Clinical Problem at a Glance

Presents the R-vs-N task, core cohort, and headline metrics.



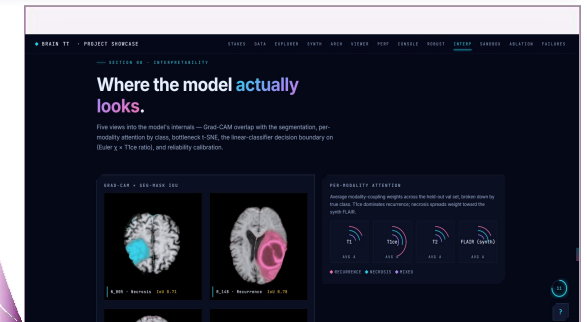
Missing-Modality Synthesis

Interactive FLAIR synthesis from available T1 / T2 signals.



Performance and Baselines

Compares BrainTT with CNN, U-Net, Transformer, and foundation baselines.



Interpretability and Audibility

Visualizes Grad-CAM, modality attention, decision boundary, and calibration.

# CONTENTS

- 1 Motivation and Values
- 2 Data Acquisition and Description
- 3 Research Questions and Methods
- 4 Experiments and Results
- 5 Timeline and Milestones

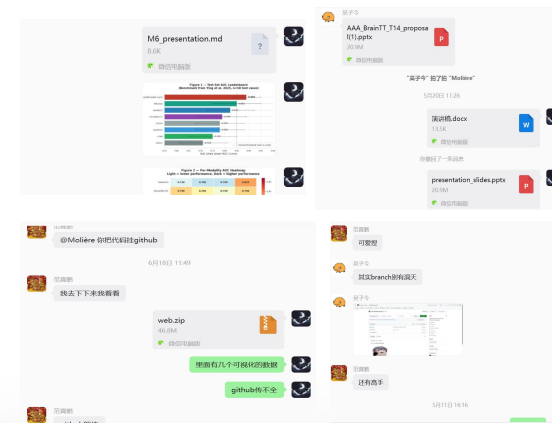
# Team and Roles

Leader: Yutong Wang

Principal Researcher  
Modeling, network design data  
processing, documentation

We collaborate through  
Wechat Group, following  
Agile developing

Each sprint leader make  
plans and deliver tasks



Zijin Wu  
ZijinWu

Zijin Wu

ML Engineer

Network design,  
training and  
testing



baigei  
24baigei

Xiaopeng Fan

Data Scientist

Data curation,  
visualization,  
conduct EDA

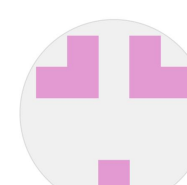


ye52  
1234asddf

Ye Wang

Research  
Scientist

Evaluation  
protocol,  
baseline testing



shangyf0528

Yunfei Shang

Product  
Manager

Documentation,  
visualization,  
testing

We have completed research experience and solid research foundation

## DIR-Net: A Disentangled Implicit Routing Network For Effective Time Series Forecasting

Jiaxue Yang<sup>a</sup>, Yutong Wang<sup>a,1</sup>, Zhongfeng Kang<sup>a,\*</sup>, Shaoyi Zong<sup>a</sup>, Shantian Yang<sup>b</sup>, Qinghua Zhao<sup>c</sup> and Zichen Song<sup>d</sup>

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### ARTICLE INFO

**Keywords:**  
Time Series Forecasting  
Temporal Product Attention  
Seasonal Pattern Capturing  
Adaptive Feature Gating  
Kolmogorov-Arnold Network

### ABSTRACT

Time series forecasting is pivotal across diverse real-world applications, including energy management, financial trading, and climate monitoring. However, prevailing forecasting architectures often homogenize the processing of trend and seasonal components or rely on predefined fusion strategies. This limits their capacity to effectively capture the heterogeneous temporal dynamics inherent in complex time series. To address these limitations, we propose the Disentangled Implicit Routing Network (DIR-Net), a novel framework that leverages disentangled dual-pathway modeling coupled with an adaptive feature routing mechanism. DIR-Net introduces a Disentangled Temporal Encoder

## MSK-Net: Multi-Scale Spatial KANs Enhanced U-Shaped Network For Explainable 3D Brain Tumor Segmentation

Yutong Wang<sup>a</sup>, Zhongfeng Kang<sup>a,\*</sup>, Xiaopeng Fan<sup>a</sup>, Zijin Wu<sup>a</sup>, Shantian Yang<sup>b</sup>, Qinghua Zhao<sup>c</sup> and Zichen Song<sup>d</sup>

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### ARTICLE INFO

**Keywords:**  
3D Medical Image Segmentation  
Kolmogorov-Arnold Networks  
Explainability  
Multi-Scale Feature Fusion  
Brain Tumor Segmentation

### ABSTRACT

Gliomas, the most common and aggressive primary brain tumors, exhibit substantial morphological complexity and tissue heterogeneity. Accurate segmentation from Magnetic Resonance Imaging (MRI) is critical for diagnosis, treatment planning, and prognosis. Despite recent advances in automated approaches, challenges persist in jointly achieving high 3D segmentation accuracy, computational efficiency, and interpretability. To address these, this study proposes MSK-Net (Multi-Scale KAN-U-Net), a novel 3D segmentation framework that incorporates the Kolmogorov-Arnold Network

## Think Globally, Act Locally: A Hierarchical Multi-Agent Framework with Retrieval-Augmented Generation

Yutong Wang<sup>a</sup>, Yunfei Shang<sup>a</sup>, Zijin Wu<sup>a</sup>, Zirui Liu<sup>a</sup>, Yina Cao<sup>b</sup> and Zhongfeng Kang<sup>a,\*</sup>

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### ARTICLE INFO

**Keywords:**  
Multi-Agent Systems  
Large Language Models  
Retrieval-Augmented Generation  
Hierarchical Coordination  
Information Fusion

### ABSTRACT

LLM-powered multi-agent systems have shown strong potential for complex task solving, yet existing flat architectures suffer from quadratic communication overhead, knowledge fragmentation across agents, and an inherent tension between centralized consistency and decentralized efficiency, all of which limit their scalability. We propose CMHA-RAG, a hierarchical multi-agent framework that *thinks globally while acting locally*. The framework couples four components: a three-tier Hierarchical Agent Topology Network (HATN) for structured task coordination, a Hierarchical RAG (H-RAG) mechanism with cascading retrieval and upward knowledge distillation, a Semantic Compression Communication Protocol (SCCP) for efficient inter-agent messaging, and an Adaptive Task Decomposition and Load Balancing (ATDLB) strategy for capability-aware resource

## KAN-Boosted Chinese Online Abuse Detection Framework with Sentiment and Toxicity Fusion through Global-Local-Differential Attention

Yutong Wang<sup>a</sup>, Zhongfeng Kang<sup>a,\*</sup>, Jiaxue Yang<sup>a</sup>, Xiaopeng Fan<sup>a</sup>, Zijin Wu<sup>a</sup>, Shantian Yang<sup>b</sup>, Qinghua Zhao<sup>c</sup> and Zichen Song<sup>d</sup>

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### ARTICLE INFO

**Keywords:**  
Online Abuse Detection  
Sentiment Analysis  
Global-Local-Differential Attention  
Kolmogorov-Arnold Networks  
Feature Fusion

### ABSTRACT

Detecting online abuse on digital platforms is a critical yet challenging task, particularly in Chinese social media, where colloquial expressions, emojis, and abstract semantics complicate automated identification. Existing large language models (e.g., OpenAI's GPT) often exhibit limited effectiveness in Chinese contexts, yielding low precision in nuanced scenarios, while smaller task-specific models struggle to capture subtle emotional cues and implicit toxicity. Moreover, most existing architectures emphasize toxicity alone, overlooking sentiment features and thereby constraining overall performance. To address these challenges, we propose Sentox-GLDA, a KAN-boosted framework for Chinese online abuse detection that fuses sentiment and toxicity features through



# Gantt Chart

Two phases, seven sub-tasks, week-level



| Tasks Decomposition |                                   | May. 2026 |   |   |   | Jun. 2026 |   |   |   | Jul. 2026 |
|---------------------|-----------------------------------|-----------|---|---|---|-----------|---|---|---|-----------|
| Task name           |                                   | 1         | 2 | 3 | 4 | 1         | 2 | 3 | 4 | 1         |
| Phase 1             | Data curation, EDA, prior design  |           |   |   |   |           |   |   |   |           |
|                     | Baseline pipeline, training infra |           |   |   |   |           |   |   |   |           |
|                     | First prior augmented varient     |           |   |   |   |           |   |   |   |           |
| Phase 2             | Prior module ablation study       |           |   |   |   |           |   |   |   |           |
|                     | Class imbalance treatment study   |           |   |   |   |           |   |   |   |           |
|                     | Interpretability analysis         |           |   |   |   |           |   |   |   |           |
|                     | Final documentation               |           |   |   |   |           |   |   |   |           |



# Limitations and Future Works

**Core limitation: scale of the held-out validation split, only 64 cases**

- The training cohort is single-institutional because it only contains patients data from Tiantan hospital. When conducting zero shot inference on Huashan data, the AUC drops.

## Future Academic Direction

**More strong baselines**

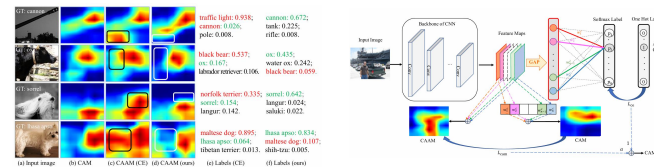
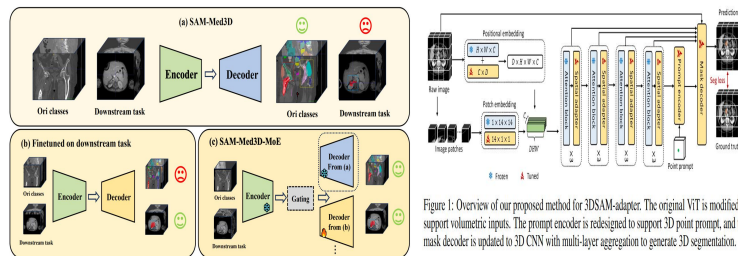
Mamba based models, RWKV based models, SAM based models……

**Method Improvement**

CAM based loss for more interpretability, tversky loss for imbalanced classes

**Academic Paper**

Complete the writing of academic paper and submit it during Sep. and Oct.



$$T(\alpha, \beta) = \frac{\sum_{i=1}^N p_{0i} g_{0i}}{\sum_{i=1}^N p_{0i} g_{0i} + \alpha \sum_{i=1}^N p_{0i} g_{1i} + \beta \sum_{i=1}^N p_{1i} g_{0i}}$$



enrich experiments and visualiztion

more robust and strong method

MICCAI, IPMI, ISBI, MIDL……



# Thanks for listening

<https://github.com/Emolier-s-hajimi/LZU-INFO442-BrainTT>.

<https://braintt.vercel.app>

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Ye Wang(ID:320230942541), Yunfei Shang(ID:320230942271)